

ROHIT KUMAR

Embedded Systems | Hardware Design | PCB Design | IoT
[LinkedIn](#) | New Delhi | Open to Relocate | Immediate Joiner

PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with hands-on experience in Embedded Systems, PCB Design, Hardware Design, and IoT. Proficient in Embedded C, Arduino, ESP32/ESP8266, PCB layout, sensor interfacing, hardware debugging, and embedded firmware development. Seeking an entry-level role in Embedded Systems, Hardware Design, PCB Design, or IoT Engineering.

CORE COMPETENCIES

Embedded Systems • PCB Design • Hardware Design • Embedded Firmware • Sensor Interfacing • Hardware Debugging • IoT • Circuit Design

TECHNICAL SKILLS

Programming: C, Embedded C, Python

Embedded Systems: Arduino, ESP32, NodeMCU (ESP8266), GPIO, UART, SPI, I²C, PWM, ADC, EEPROM, Sensor Interfacing, Embedded Firmware Development

Hardware & PCB Design: Circuit Design, Schematic Design, PCB Design, PCB Layout (EasyEDA), Circuit Prototyping, PCB Assembly, Soldering, Hardware Debugging, Circuit Testing, Troubleshooting

IoT & Communication: ThingSpeak, Wi-Fi, Bluetooth (HC-05), IoT Integration, Sensors & Actuators

Software Tools: Arduino IDE | EasyEDA | LTspice | Xilinx Vivado

PROFESSIONAL EXPERIENCE

STEM Innovation Engineer - StemMantra Pvt. Ltd.

Sep 2025 - Nov 2025

- Developed embedded prototypes using Arduino, ESP32, and ESP8266 microcontrollers.
- Integrated sensors and actuators using UART, SPI, I²C, GPIO, ADC, and PWM interfaces.
- Designed, assembled, and tested embedded electronic circuits and hardware prototypes.
- Performed PCB soldering, troubleshooting, hardware debugging, and prototype validation.
- Built IoT-based applications integrated with ThingSpeak for real-time monitoring.
- Supported embedded firmware development, testing, and technical demonstrations.

VLSI Design Intern - Sure Trust

Sep 2024 – Jun 2025

- Designed RTL modules using Verilog HDL.
- Gained exposure to SystemVerilog and basic UVM verification methodology.
- Performed simulation using Xilinx Vivado.
- Gained understanding of ASIC design flow and FPGA implementation.

PROJECTS

IoT Smart Air Purifier

Technologies: ESP8266, MQ135, ThingSpeak

- Designed an IoT-enabled air purification system using ESP8266 and environmental sensors.
- Automated fan speed control based on real-time air quality measurements.
- Integrated ThingSpeak cloud platform for live monitoring and data visualization.
- Performed firmware testing, hardware debugging, and system validation.

IoT Health Monitoring System

Technologies: ESP8266, MAX30100, AD8232 ECG

- Developed a real-time health monitoring system for heart rate, SpO₂, and ECG measurement.
- Integrated multiple biomedical sensors using I²C and analog interfaces.
- Uploaded live physiological data to ThingSpeak cloud for remote monitoring.
- Improved system accuracy through hardware testing and sensor calibration.

RFID-Based Door Lock System

Technologies: Arduino, RFID, EEPROM

- Developed a secure RFID-based access control system using Arduino.
- Implemented EEPROM-based card authentication and access management.
- Controlled door locking mechanism using relay and motor driver.
- Performed hardware integration, testing, and debugging.

EDUCATION

Bachelor of Technology (Electronics & Communication Engineering)

Guru Jambheshwar University of Science and Technology
2021 – 2025 | CGPA: 6.23/10

ACHIEVEMENTS

- 3rd Prize – Startup for Developing Entrepreneurial Qualities (University Level)
- Selected for State-Level Startup Competition

LEADERSHIP

Joint Department Coordinator - Industrial Interaction Club (JDIIC)

- Organized technical workshops, industry interaction sessions, and departmental events.

NSS Volunteer

- Participated in technical and community outreach programs.